

Assignment 2: Sanskrit-to-English Neural Machine Translation

1. Introduction

This report explains the development of a Neural Machine Translation (NMT) system designed to translate text from Sanskrit to English. Sanskrit is a highly inflected, morphologically rich language, making it a challenging low-resource NLP task. The objective of this project is to implement a sequence-to-sequence (Seq2Seq) architecture from scratch that effectively bridges the syntactic and semantic gap between the two languages while adhering to strict parameter and inference time efficiency constraints.

2. Architecture

The system utilizes a custom PyTorch Transformer Seq2Seq architecture rather than a recurrent neural network to better capture long-range dependencies in the text.

- **Encoder & Decoder:** The model comprises 3 encoder layers and 3 decoder layers, utilizing a multi-head attention mechanism with 8 heads. The embedding dimension is set to 256, and the feed-forward network dimension is 512.
- **Decoding Strategy:** While Beam Search is a standard approach for NMT, this implementation utilizes an **Optimized Parallel Batch Greedy Decoder**. This design choice was made to heavily prioritize computational efficiency and inference speed, allowing the model to decode batches in parallel on the GPU while capping maximum sequence length to prevent hallucination loops.

3. Training Procedure

- **Preprocessing & Tokenization:** Standard whitespace tokenization fails on the complex morphology of Sanskrit. Therefore, a Hugging Face ByteLevelBPETokenizer was trained from scratch on the parallel training corpus. By operating at the byte level, the tokenizer natively handles spaces and unknown characters without complex pre-processing rules, effectively managing out-of-vocabulary (OOV) tokens. The vocabulary size was explicitly constrained to 3,000 for both source and target languages to maintain a lightweight embedding layer.
- **Optimization:** The model was trained for 35 epochs using a batch size of 64. The AdamW optimizer was utilized with a weight decay of 1×10^{-4} .
- **Regularization & Stability:** To prevent the model from becoming overconfident on the small 10,000-sentence training set, a dropout rate of 0.20 was applied. Furthermore, CrossEntropyLoss was implemented with a Label Smoothing factor of 0.1. Finally, a custom learning rate scheduler was introduced, utilizing a 10% Linear Warmup phase followed by Cosine Annealing, which proved critical in preventing early gradient divergence.

4. Results

The model was evaluated against the private test set. Efficiency metrics prioritize rapid execution and a minimal memory footprint.

- **BLEU Score:** 0.06882
- **BERTScore F1 (Rescaled):** 0.21698
- **Inference Time:** 8.2452 seconds (Avg: ~8.2 ms per sentence)
- **Total Parameters:** 6,261,688 (6.262M)

Disclosure regarding Pre-trained Models: The core NMT model was built and trained entirely from scratch. The pre-trained Hugging Face model roberta-large was downloaded and utilized *exclusively* under the hood by the bert-score library to compute the evaluation metrics, in accordance with the assignment instructions.

5. Translation Examples & Error Analysis

(Note: An evaluation of model predictions highlights specific failure modes typical of low-resource NMT).

- **Example 1: Semantic Approximation (Lexical Substitution)**
 - **Source:** बालः युष्मासु विश्वासं करोति ।
 - **Reference:** The boy trusts you all.
 - **Prediction:** "child has love in you all."
 - **Analysis:** The model successfully captured the core semantics and structure of the sentence, contributing to a positive BERTScore. It correctly mapped "बालः" to "child/boy" and "युष्मासु" to "in you all". However, it substituted "विश्वासं" (trust/faith) with a semantically adjacent word, "love", resulting in an exact n-gram penalty (BLEU).
- **Example 2: Pronoun and Morphological Misalignment**
 - **Source:** बालः त्वयि प्रेमं प्रदर्शयति ।
 - **Reference:** The boy shows love to you.
 - **Prediction:** "boy displays affection in them."
 - **Analysis:** The model generated a highly accurate semantic translation for the primary subject and verb ("boy displays affection" for "बालः ... प्रेमं प्रदर्शयति"). However, it failed to correctly align the Sanskrit locative pronoun "त्वयि" (in/to you), mapping it instead to the third-person plural "in them".
- **Example 3: Entity Substitution and Precision Loss**
 - **Source:** XAMPP 5.5.19 द्वारा प्राप्तानि Apache, MySQL तथा PHP च,
 - **Reference:** Apache, MySQL and PHP obtained through XAMPP 5.5.19,
 - **Prediction:** "apache, mariadb and php obtained through xampp 5.5.1."

- **Analysis:** The model successfully translated the technical structure of the sentence. Interestingly, it substituted "MySQL" with "mariadb", strongly indicating a bias in the training corpus where these database terms appear in similar contexts. Furthermore, it suffered a precision loss on the version number (truncating 5.5.19 to 5.5.1).
- **Example 4: The Hallucination Loop (Short Context)**
 - **Source:** अपरं द्वितीयमुद्रायां तेन मोचितायां द्वितीयस्य प्राणिन आगत्य पश्येति वाक् मया श्रुता।
 - **Reference:** And when he had opened the second seal, I heard the second beast say, Come and see.
 - **Prediction:** "and when he had spoken, he had spoken, he had spoken, he had received him."
 - **Analysis:** This is a classic failure mode of greedy decoding. The decoder successfully initiated the translation ("and when he had...") but failed to map the complex morphological inflections of the subsequent Sanskrit words. Without beam search to prune low-probability paths, the model fell into an unrecoverable repetition loop.
- **Example 5: Catastrophic Failure on Long Sequences**
 - **Source:** सौविधाय प्रयुज्यमानानि वस्तूनि यथा घटीयन्त्रं लेखनी दूरवाणी शीतयन्त्रं(फ्रिज) दूरदर्शनं यानम इत्यादीनां निर्माणायापि विविधानाम आधुनिकपदार्थानामुपयोगः भवति।
 - **Reference:** Various modern materials are also used to manufacture objects used for convenience, such as a clock, pen, telephone, refrigerator, television, vehicle, etc.
 - **Prediction:** "the syntax for example, you will be able to get more than or more than or more than or more than or more than or more than or more than or more than or more than or more than or more than or more than or more than or more than or deeds."
 - **Analysis:** The model completely failed to translate this long, morphologically dense sentence. The presence of compound words and list items caused the attention mechanism to lose context entirely, resulting in a degenerate output consisting only of high-frequency stop-words and looping syntax.

6. Experiments & Discussion

Developing this system highlighted the stark contrast between representation tasks and generation tasks in NLP. Previous architectural implementations focusing on semantic embeddings for parallel text (using models like Sentence Transformers) rely on projecting inputs into a shared vector space, which is highly effective for identifying similarity. However, this NMT assignment requires true sequence-to-sequence generation, which is significantly more complex. The model must not only map the semantic meaning but also independently reconstruct grammatically sound English syntax from scratch.

The primary limitation encountered was the small dataset size (10,000 pairs). Transformers are notoriously data-hungry and lack inherent inductive biases for sequence translation. Early experiments showed severe overfitting. Implementing Byte-Level BPE, aggressive label smoothing, and a learning rate warmup were necessary design choices to stabilize training and extract a

meaningful BLEU and BERTScore from such a constrained corpus. Future iterations would greatly benefit from integrating a Beam Search decoding strategy to prune grammatically incorrect branches during inference.

7. References

1. Papineni, K., et al. (2002). BLEU: a method for automatic evaluation of machine translation. *ACL*.
2. Zhang, T., et al. (2019). BERTScore: Evaluating Text Generation with BERT. *ICLR*.
3. Hugging Face (2024). *tokenizers* Library.
4. Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. *NeurIPS*.